CONCEPT, DESIGN AND OPERATION (RUNNING) OF A WEB PAGE

Concept of a Web Page

A web page is a digital document that is created using web technologies and displayed on the internet through a web browser. It forms the basic unit of a website and is designed to present information in text, images, audio, video, or interactive formats. A web page is accessed through a unique address known as a Uniform Resource Locator (URL) and is retrieved from a web server when requested by a user.

From an information science perspective, a web page functions as a platform for information storage, dissemination, and retrieval in the electronic environment. Libraries use web pages to provide access to catalogues, databases, institutional repositories, library guides, announcements, and user education materials. Unlike printed documents, web pages are dynamic, meaning they can be updated easily and accessed by multiple users simultaneously from different locations.

A web page operates within the framework of the World Wide Web and relies on internet connectivity, browsers, and web servers to function effectively.

Components of a Web Page

A web page is made up of several interconnected components that work together to display content properly. At the core of a web page is **content**, which includes text, images, videos, links, and other information elements. This content is structured using markup language so that browsers can interpret and display it correctly.

Another important component is **layout and structure**, which determines how content appears on the screen. Layout controls the arrangement of headings, paragraphs, images, menus, and other elements. **Navigation** is also a key component, allowing users to move from one section or page to another through hyperlinks and menus.

In addition, web pages often include **interactive elements** such as forms, search boxes, and buttons, which enable users to interact with the page. Behind the scenes, web pages may also contain scripts and style definitions that enhance functionality and visual presentation.

Concept of Web Page Design

Web page design refers to the process of planning, creating, and organising the visual and functional aspects of a web page to ensure effective communication of information. It combines creativity with technical knowledge to produce pages that are attractive, functional, and user-friendly.

In web page design, emphasis is placed on **usability**, **accessibility**, and **aesthetics**. A well-designed web page allows users to find information easily without confusion. For libraries

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and information centres, web page design is particularly important because users depend on the library website to access resources, services, and support.

Web page design is not limited to appearance alone; it also considers how information is structured, how quickly pages load, and how well the page works across different devices such as computers, tablets, and mobile phones.

Principles of Effective Web Page Design

Effective web page design is guided by certain principles that ensure clarity and functionality. One major principle is **simplicity**, which requires avoiding unnecessary elements that may distract users. Another principle is **consistency**, ensuring uniform fonts, colours, and layouts across pages.

Readability is also crucial. Text must be clear, properly spaced, and easy to read. **Navigation clarity** ensures that users can easily move through the page and locate desired information. **Responsiveness** ensures that the web page adjusts automatically to different screen sizes.

In the library context, accessibility is especially important. Web pages should be designed so that users with disabilities can access information through screen readers, alternative text for images, and simple navigation structures.

Tools and Languages Used in Web Page Design

Web pages are created using a combination of languages and tools. **HTML (Hypertext Markup Language)** is used to structure the content of a web page. It defines headings, paragraphs, images, links, and other elements.

CSS (Cascading Style Sheets) is used to control the appearance of the web page, including colours, fonts, spacing, and layout. CSS separates content from design, making web pages easier to manage and update.

JavaScript is used to add interactivity to web pages, such as form validation, dynamic content loading, and user interaction features. Together, HTML, CSS, and JavaScript form the foundation of modern web page design.

Web design tools such as text editors, content management systems, and web development platforms are also used to simplify the design process.

Concept of Running a Web Page

Running a web page refers to the process of making a web page functional and accessible to users. This involves testing, hosting, and accessing the page through a web browser. A web

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page is considered "running" when it can be successfully loaded, displayed, and interacted with by users.

There are two major contexts in which a web page can be run: **locally** and **online**. Running a web page locally involves opening the web page files directly on a computer using a browser. Running a web page online involves uploading the files to a web server so that users can access the page through the internet.

How to Run a Web Page Locally

To run a web page locally, the designer first creates the web page files using HTML, CSS, and possibly JavaScript. These files are saved on a computer. When the HTML file is opened using a web browser such as Chrome or Firefox, the browser reads the code and displays the web page.

Running a web page locally is useful for testing and learning purposes. It allows designers to preview changes instantly without internet access. However, local pages are only accessible on the computer where they are stored.

How to Run a Web Page Online

Running a web page online requires a **web server**. A web server is a computer system that stores web page files and delivers them to users when requested through the internet. To run a web page online, the designer uploads the web page files to a hosting server using internet services.

Once uploaded, the web page is assigned a domain name or URL. When users type this URL into a browser, the browser sends a request to the server, which then delivers the web page to the user's device.

Libraries use web servers to host their websites, online catalogues, databases, and digital repositories, enabling users to access services remotely.

Importance of Web Pages in Libraries and Information Centres

Web pages play a critical role in modern library services. They serve as gateways to electronic resources, provide access to OPACs and databases, support virtual reference services, and disseminate information to users.

Through well-designed web pages, libraries enhance visibility, improve user engagement, and support teaching and research. A functional web page strengthens the library's relevance in the digital age.

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Conclusion

The concept, design, and running of a web page are fundamental aspects of information service delivery in the electronic environment. A web page is more than a digital document; it is a platform for information access, communication, and interaction. Effective web page design ensures usability, accessibility, and clarity, while proper running of a web page ensures that information is available to users anytime and anywhere.

For library and information professionals, understanding web page design and operation is essential for delivering efficient, modern, and user-centred library services.

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